

Elementary Linear Algebra Notes Part I

MATH 1890

Spring 2025

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Outline

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- Vector and Matrix Equations
- Linear Independence and Solution Sets of Linear Systems
- Introduction to Linear Transformations

2 Matrix Algebra

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- Subspaces of $\mathbb{R}^2$
- Dimension and Rank

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- Properties of Determinants
- Cramer's Rule, Volume, and Linear Transformations

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- Vector Spaces and Subspaces
- Null Spaces and Column Spaces
- Linearly Independent Sets and Bases

Systems of Linear Equations

- A **real number** is a number that can be used to measure a continuous one-dimensional quantity such as a distance, duration or temperature. In short, any number you encounter in real world measurements or transactions is a real number. For example; $4, 2.8, \pi, 1.333\dots, -\frac{2}{7}, e, \sqrt{2}$, etc.

Consider a set of n variables $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$.

A **linear equation** is a combination of scalar multiples of the n variables to yield an output. This can be written in the form

$$a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \dots + a_n\mathbf{x}_n = b \quad (1)$$

where b and the scalars $a_1, a_2, \dots, a_n$ are real numbers.

A **system of linear equation** collection of equations of the form in equation (1).

Solutions to Systems of Linear Equations

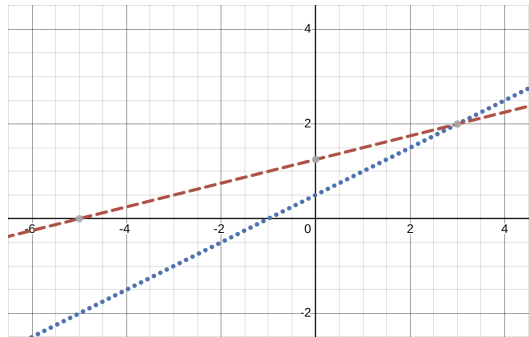
The **solution to a linear system** is the point or points that satisfy each of the linear equations within the system. There can be one, or many, or none. If it has one or many solutions, the system is **consistent**. Otherwise it is inconsistent.

The linear system

$$-2x_1 + 4x_2 = 2 \quad (2)$$

$$-x_1 + 4x_2 = 5 \quad (3)$$

has a unique solution, $x_1 = 3$, $x_2 = 2$ because the two lines intersect at one and only one point.



Solutions to Systems of Linear Equations

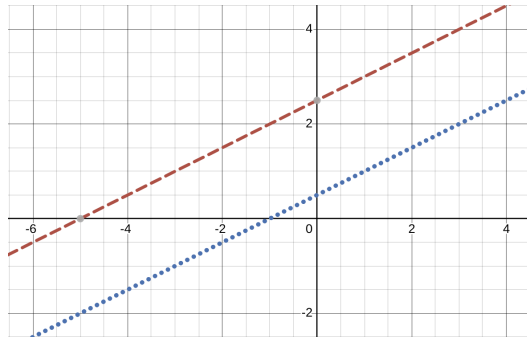
The **solution to a linear system** is the point or points that satisfy each of the linear equations within the system. There can be one, or many, or none. If it has one or many solutions, the system is **consistent**. Otherwise it is inconsistent.

The linear system

$$-2x_1 + 4x_2 = 2 \quad (2)$$

$$-x_1 + 2x_2 = 5 \quad (3)$$

has no solution because the two lines are parallel and never intersect.



Solutions to Systems of Linear Equations

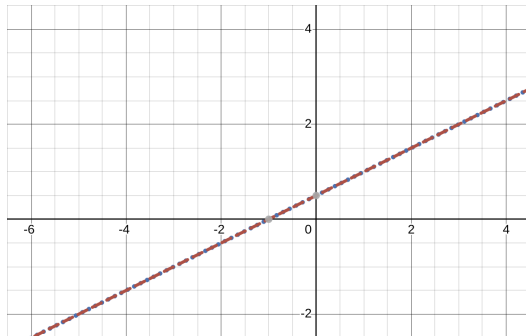
The **solution to a linear system** is the point or points that satisfy each of the linear equations within the system. There can be one, or many, or none. If it has one or many solutions, the system is **consistent**. Otherwise it is inconsistent.

The linear system

$$-2x_1 + 4x_2 = 2 \quad (2)$$

$$-x_1 + 2x_2 = 1 \quad (3)$$

has infinitely many solutions because the lines intersect everywhere (at infinite number of points).



Solving Systems of Equations

A linear system may be represented as a matrix and solved using a series of the following row operations:

- **Replacement:** Replace one row by the sum of itself and a multiple of another row.
- **Interchange:** Interchange two rows.
- **Scaling:** Multiply all entries in a row by a nonzero real number.

For instance, the linear system

$$-2x_1 + 4x_2 + 6x_3 = 2$$

$$-x_1 + 7x_2 + x_3 = 5$$

$$3x_1 - 7x_2 + 2x_3 = 1$$

can be written as

$$\left[\begin{array}{ccc|c} -2 & 4 & 6 & 2 \\ -1 & 7 & 1 & 5 \\ 3 & -7 & 2 & 1 \end{array} \right]$$

Complete this row operation so that you have an upper triangular matrix (all values below the main diagonal are zeros) as your **practice**.

Solving Systems of Equations

We then perform the following row operations:

$$\left[\begin{array}{ccc|c} -2 & 4 & 6 & 2 \\ -1 & 7 & 1 & 5 \\ 3 & -7 & 2 & 1 \end{array} \right] - \frac{1}{2}R_1 \rightarrow R_1 \left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ -1 & 7 & 1 & 5 \\ 3 & -7 & 2 & 1 \end{array} \right]$$

$$R_1 + R_2 \rightarrow R_2 \left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ 0 & 5 & -2 & 4 \\ 3 & -7 & 2 & 1 \end{array} \right] - 3R_1 + R_3 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ 0 & 5 & -2 & 4 \\ 0 & -1 & 11 & 4 \end{array} \right]$$

Complete this row operation so that you have an upper triangular matrix (all values below the main diagonal are zeros) as your **practice**.

Question Q1 Given the linear system

$$x_1 - 2x_2 + x_3 = 0$$

$$2x_2 - 8x_3 = 8$$

$$-4x_1 + 5x_2 + 9x_3 = -9$$

- a Rewrite the system in matrix notation.
- b Using elementary row operations only, reduce the system to an upper triangular matrix.
- c Using the upper triangular matrix, solve for x_1 , x_2 , and x_3 .
Note: *for an inconsistent system, you will not be able to solve the system*
- d Is the system consistent?

Homework 1

Question Q2 Given the linear system

$$x_2 - 4x_3 = 8$$

$$2x_1 - 3x_2 + 2x_3 = 1$$

$$5x_1 - 8x_2 + 7x_3 = 1$$

- a Rewrite the system in matrix notation.
- b Using elementary row operations only, reduce the system to an upper triangular matrix.
Hint: *You may need to swap rows to achieve this.*
- c Using the upper triangular matrix, solve for x_1 , x_2 , and x_3 . **Note:** *for an inconsistent system, you will not be able to solve the system.*
- d Is the system consistent?

Row Reduction and Echelon Forms

A rectangular matrix is in **echelon form** if it has the following properties:

- Any row with all zeros should be moved to the last row.
- Each leading entry should have only zeros below it in that column.

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare \end{bmatrix}$$

Note: a leading entry is the first nonzero term in a row.

Theorem (Uniqueness of the Reduced Echelon Form)

Each matrix is equivalent to one and only one reduced echelon matrix.

Row Reduction and Echelon Forms

A rectangular matrix is in **echelon form** if it has the following properties:

- Any row with all zeros should be moved to the last row.
- Each leading entry should be **1** and have only zeros below and above it in that column.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Note: a leading entry is the first nonzero term in a row.

Theorem (Uniqueness of the Reduced Echelon Form)

Each matrix is equivalent to one and only one reduced echelon matrix.

Example

Row reduce the matrix below to **reduced** echelon form

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}.$$

Echelon form:

$$A = \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solving Linear Systems with Row Reductions

Suppose your row reduction yields the following, we solve using back substitution;

$$\left[\begin{array}{ccc|c} 1 & -2 & -3 & -1 \\ 0 & 1 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

We say that x_3 is a free variable, and hence

$$x_3 = x_3,$$

$$x_2 - 2x_3 = 4$$

$$\text{so, } x_2 = 4 + 2x_3, \text{ and}$$

$$x_1 - 2x_2 - 3x_3 = -1$$

$$\text{so, } x_1 = -1 + 2x_2 + 3x_3.$$

$$\begin{aligned} \text{Thus, } x_1 &= -1 + 2(4 + 2x_3) + 3x_3 \\ &= 7 + 7x_3. \end{aligned}$$

Homework 2

Determine whether each of the following matrices is in *echelon form*, *reduced echelon form*, or *neither*. Justify your answers.

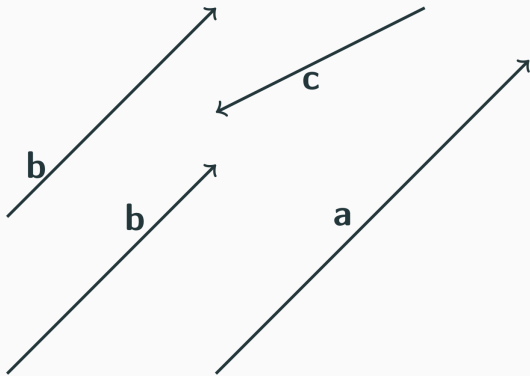
Given Matrices:

$$A_1 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$A_4 = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_5 = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 4 \end{bmatrix}, \quad A_6 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Vectors

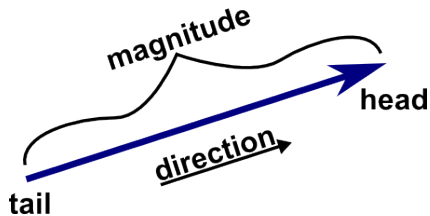
Linear algebra is a branch of mathematics that focuses on the study of **vectors**, the space within which they exist (**vector spaces**) as well as their characteristics and operations that can be performed on them. Provides a mathematical framework for analyzing systems of linear equations and their representations in higher dimensions.



Another View is that Vectors are **ordered lists of numbers**. Consider the model for the price of gold.

$$\begin{bmatrix} 1\text{oz} \\ \$1200 \end{bmatrix} \neq \begin{bmatrix} \$1200 \\ 1\text{oz} \end{bmatrix}$$

Vectors have **magnitude** and **direction** (arrows pointing in

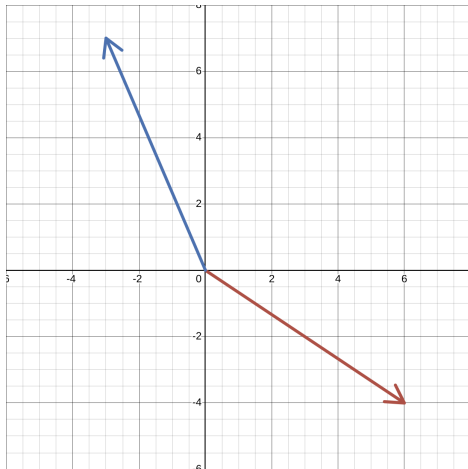


In this course when we say vectors we will be referring to a directed line segment sitting at the origin.

- The two numbers (coordinates) are a pair of numbers that tell you how to get from the tail(origin) to the head(tip)

Figure: https://mathinsight.org/vector_introduction

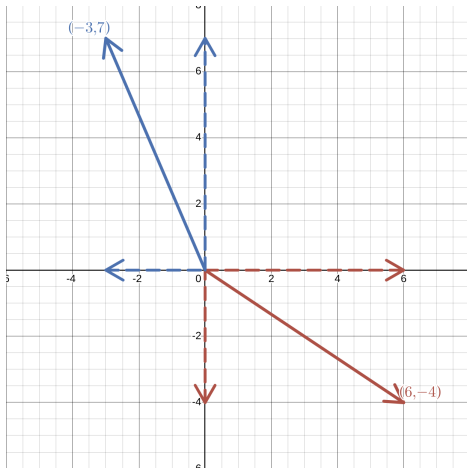
Vectors



Consider the vectors $\begin{bmatrix} -3 \\ 7 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ -4 \end{bmatrix}$

- The first number tells you how far to walk on the x -axis. **Positive** numbers mean **right**, and **negative** numbers mean **left**.
- The second number tells you how far to walk parallel to the y -axis. **Positive** numbers mean **up**, and **negative** numbers mean **down**.
- Every pair of numbers is associated with only one vector (arrow), and every vector (arrow) is associated with only one pair of numbers.

Vectors



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Addition of Vectors and Scalar Multiplication

Since vectors are ordered (component-wise), addition is also component-wise.

- **Addition:** Subtraction is also a form of addition

$$\begin{bmatrix} -3 \\ 7 \end{bmatrix} + \begin{bmatrix} 6 \\ -4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

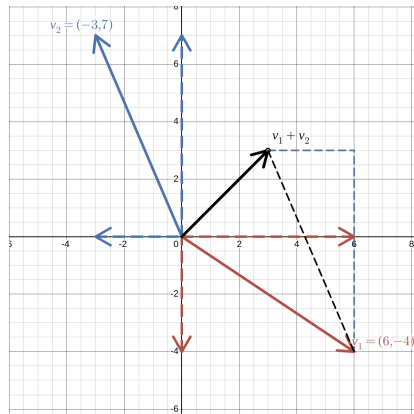


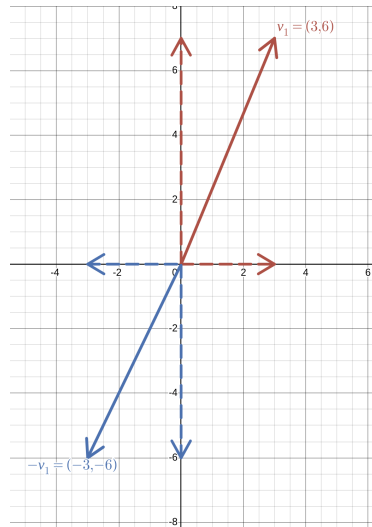
Figure: Vector Addition

Addition of Vectors and Scalar Multiplication

Since vectors are ordered (component-wise), addition is also component-wise.

- **Scalar Multiplication**

$$-v_1 = - \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} -3 \\ -6 \end{bmatrix}$$



Addition of Vectors and Scalar Multiplication

Since vectors are ordered (component-wise), addition is also component-wise.

- **Scalar Multiplication**

$$3v_2 = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

Note: a vector can have more than 2 dimensions.

$\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ is a 3-dimensional vector, and $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ is n-dimensional v

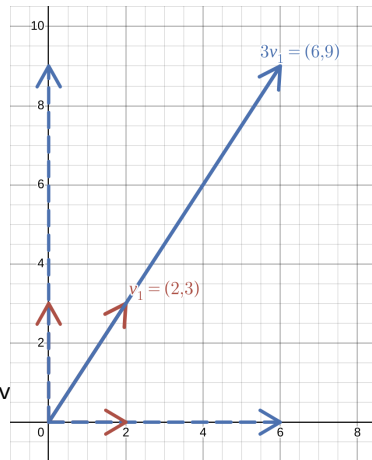


Figure: Scalar Multiplication

Special Vectors (Basis)

Consider the special vectors $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Observe that the vectors $\begin{bmatrix} -3 \\ 7 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ -4 \end{bmatrix}$ can be written as a sum and scalar multiplication of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$. $\begin{bmatrix} -3 \\ 7 \end{bmatrix} = -3\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$, and $\begin{bmatrix} 6 \\ -4 \end{bmatrix} = 6\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$.

All the vectors you can reach using only vector addition and scalar multiplication is called the **span** of these special vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$. This is also known as the **linear** combination of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ (Desmos illustration <https://www.desmos.com/calculator/apimvn3ln9>)

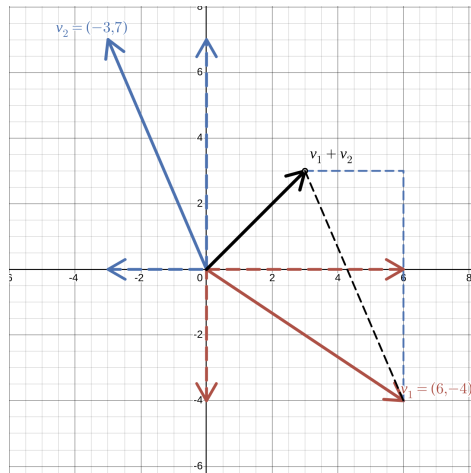


Figure: Vector Addition

Algebraic Properties of Vectors in $\mathbb{R}^n$ and Matrix Equations

- **Commutative Property:**

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

- **Associative Property:**

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

- **Additive Identity:** $\mathbf{u} + \mathbf{0} = \mathbf{u}$

- **Additive Inverse:** $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

- **Distributive Property:**

$$r(\mathbf{u} + \mathbf{v}) = r\mathbf{u} + r\mathbf{v}$$

Where $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors. r is a scalar

A vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n = \mathbf{w}$$

has the same solution set as the linear system whose augmented matrix is

$$[\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n \ \mathbf{w}] .$$

In the vector equation, we say that $\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$.

Definition (*Span* of a Set of Vectors)

The *span* of a set of vectors $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p \in \mathbb{R}^n$ is the set of all possible linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p$.

That is, if a vector $\mathbf{b}$ is in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p\}$, then $\mathbf{b} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$.

Matrix Multiplication

Here are some examples to illustrate multiplication of matrices and vectors with different dimensions, showcasing various matrix equations:

Example 1: Multiplication of a Matrix and a Column Vector

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Matrix Multiplication

Matrix dimensions: A 2×2 , $\mathbf{v}$ 2×1

The product is:

$$A\mathbf{v} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \cdot 5 + 3 \cdot 6 \\ 4 \cdot 5 + 1 \cdot 6 \end{bmatrix}$$

Example 2: Multiplication of Two Matrices**

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

Matrix Multiplication

Matrix dimensions: B 2×3 , C 3×2 The product is:

$$BC = \begin{bmatrix} \textcolor{red}{1} & \textcolor{red}{2} & \textcolor{red}{3} \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} \textcolor{blue}{7} & \textcolor{cyan}{8} \\ \textcolor{blue}{9} & \textcolor{cyan}{10} \\ \textcolor{blue}{11} & \textcolor{cyan}{12} \end{bmatrix} = \begin{bmatrix} \textcolor{red}{1} & \textcolor{red}{2} & \textcolor{red}{3} \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} \textcolor{blue}{7} & \textcolor{cyan}{8} \\ \textcolor{blue}{9} & \textcolor{cyan}{10} \\ \textcolor{blue}{11} & \textcolor{cyan}{12} \end{bmatrix}$$

Example 3: Multiplication Involving a Row Vector and a Matrix

$$\mathbf{u} = [1 \quad 2 \quad 3], \quad D = \begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}$$

Matrix Multiplication

Matrix dimensions: u 1×3 , D 3×2 The product is:

$$uD = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 2 \cdot 6 + 3 \cdot 8 & 1 \cdot 5 + 2 \cdot 7 + 3 \cdot 9 \end{bmatrix} = \begin{bmatrix} 40 & 46 \end{bmatrix}$$

Example 4: Non-Conformable Matrices

$$E = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 4 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Matrix Multiplication

Matrix dimensions: E 2×3 , F 2×2

You cannot multiply EF because the number of columns in E does not match the number of rows in F . However, if their roles are reversed (FE), multiplication is possible, resulting in another 2×3 matrix.

Matrix Equations

Recall: A vector equation

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \cdots + x_n \mathbf{v}_n = \mathbf{w}. \quad (4)$$

Equation (4) may be rewritten (with columns as vectors) as

$$\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{w}$$
$$\begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$
$$A\mathbf{x} = \mathbf{w}.$$

Matrix Equation $A\mathbf{x} = \mathbf{w}$

Definition

Matrix Equation If A is an $m \times n$ matrix with columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, and if $\mathbf{x} \in \mathbb{R}^n$, then the product of A and $\mathbf{x}$, is denoted by $A\mathbf{x}$, **is the linear combination of the columns** of A using the corresponding entries in $\mathbf{x}$ as weights; i.e.,

$$A\mathbf{x} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n$$

That is, we view a linear combination of vectors as the product of a matrix and a vector.

Formulating Matrix equations

For example consider the system of linear equations,

$$\begin{aligned}x_1 + 2x_2 - x_3 &= 4 \\ -5x_2 + 3x_3 &= 1\end{aligned}$$

The corresponding Matrix equation is

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

which is equivalent to the vector equation

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Solution to Matrix Equations

Theorem

Existence of Solutions The equation $A\mathbf{x} = \mathbf{w}$ has a solution if and only if $\mathbf{w}$ is a linear combination of the columns of A .

How is this related to the *Span*?

Lemma

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent;

- (a) For each $\mathbf{w} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{w}$ has a solution.
- (b) Each $\mathbf{w} \in \mathbb{R}^m$ is a linear combination of the columns of A .
- (c) The columns of A span $\mathbb{R}^m$.
- (d) A has a pivot in every row after row reduction.

Example 1

$$A = \begin{bmatrix} 1 & -3 & -4 \\ -3 & 2 & 6 \\ 5 & -1 & -8 \end{bmatrix}$$

- Does the equation $A\mathbf{x} = \mathbf{b}$ have a solution for all possible $\mathbf{b} \in \mathbb{R}^3$?
- In other words, do the columns of A span $\mathbb{R}^3$?
- Alternatively, can any vector, $\mathbf{b} \in \mathbb{R}^3$ be written as a linear combination of the 3 columns of A ?

If it does, write out the form of each $\mathbf{b}$.

If not, describe the set of all $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ does have a solution.

Examples

Solution: according to the lemma in the previous page, each $A\mathbf{x} = \mathbf{b}$ will have a solution for each $\mathbf{b} \in \mathbb{R}^3$ if A has a pivot in every row. So perform a row reduction to get:

$$\begin{bmatrix} 1 & -3 & -4 \\ 0 & -7 & -6 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since row 3 does not have a pivot (no nonzero number in the row), we conclude that $A\mathbf{x} = \mathbf{b}$ does not have a solution for all $\mathbf{b} \in \mathbb{R}^3$.

To find the form of $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$, use row reductions with the b 's in an augmented matrix form.

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & b_1 \\ -3 & 2 & 6 & b_2 \\ 5 & -1 & -8 & b_1 \end{array} \right] \begin{array}{l} R_2 + 3R_1 \rightarrow R_2 \\ R_3 - 5R_1 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & b_2 + 3b_1 \\ 0 & 14 & 12 & b_3 - 5b_1 \end{array} \right]$$

Examples

$$R_3 + 2R_2 \rightarrow R_3 \quad \left[\begin{array}{ccc|c} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & b_2 + 3b_1 \\ 0 & 0 & 0 & 2b_2 + b_1 + b_3 \end{array} \right]$$

This matrix equation only has a solution if the entire 3rd row is 0. That is, $2b_2 + b_1 + b_3 = 0$ otherwise the system will be inconsistent. So, $b_3 = -b_1 - 2b_2$.

$$\begin{aligned} \text{Thus, } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \\ -b_1 - 2b_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ 0 \\ -b_1 \end{bmatrix} + \begin{bmatrix} 0 \\ b_2 \\ 2b_2 \end{bmatrix} = b_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + b_2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \\ \mathbf{b} &= \left\{ b_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + b_2 \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \quad \text{for } b_1, b_2 \in \mathbb{R} \right\}. \end{aligned}$$

Example 2

Let $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -3 \\ 2 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$. Do these 3 vectors span $\mathbf{R}^3$?

Solution: Yes! Reduce to echelon form and observe that each row has a pivot. So, by the previous lemma, the columns span $\mathbf{R}^3$.

Example 3

Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$. Do these 3 vectors span $\mathbb{R}^4$?

Solution: No! Reduce to echelon form and observe that the last row has no a pivot. So, by the previous lemma, the columns do not span $\mathbb{R}^4$.

Example 4 Construct a 3×3 matrix, A and vectors $\mathbf{w}, \mathbf{b} \in \mathbb{R}^3$ such that $A\mathbf{x} = \mathbf{b}$ has a solution, but $A\mathbf{x} = \mathbf{w}$ does not have a solution.

Solution: You can use the matrix in example 1. Then choose a vector that meets the condition of $\mathbf{b}$, and another vector that does not meet that condition.

Question 1

If A is an 3×2 matrix, and $\mathbf{u}, \mathbf{w} \in \mathbb{R}^2$ are vectors, show that $A(\mathbf{u} + c\mathbf{w}) = A\mathbf{u} + cA\mathbf{w}$, where c is a scalar.

That is, $A = [\mathbf{v}_1 \ \mathbf{v}_2]$ and $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$.

Question 2

Let $A = \begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$.

Show that the equation $A\mathbf{x} = \mathbf{b}$ does not have a solution for all possible $\mathbf{b}$, and describe the set of all $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has a solution. In other words, do the columns of A span $\mathbb{R}^2$?

Linear Independence and Solution Sets of Linear Systems I

Recall: (The *Span* of a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, is the set of all the possible linear combinations, $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3$).

Thus, if we can obtain $\mathbf{v}_3$ by a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, then $\mathbf{v}_3$ is **linearly dependent** on $\mathbf{v}_1$ and $\mathbf{v}_2$. Otherwise, we say $\mathbf{v}_3$ is **linearly independent** of $\mathbf{v}_1$ and $\mathbf{v}_2$.

If linear independence applies for all the elements within the set, we say S is linearly independent.

Example 1 Determine if the following homogeneous system has a nontrivial solution. Then describe the solution set.

$$3x_1 + 5x_2 - 4x_3 = 0$$

$$-3x_1 - 2x_2 + 4x_3 = 0$$

$$6x_1 + x_2 - 8x_3 = 0$$

Is $\begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix}$ as solution vector?

Example 2 Determine the solution of the following homogeneous system

$$10x_1 - 3x_2 - 2x_3 = 0.$$

Write the solution as a parametric vector equation.

A linear system is **homogeneous** if it can be written as $A\mathbf{x} = \mathbf{0}$, where A is an $m \times n$ matrix.

Linear Independence and Solution Sets of Linear Systems Part II

Theorem

Suppose that the equation $A\mathbf{x} = \mathbf{b}$ is consistent for some given $\mathbf{b}$, and let $\mathbf{p}$ be a solution. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors of the form $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$, where $\mathbf{v}_h$ is any solution of the homogeneous solution $A\mathbf{x} = \mathbf{0}$.

That is, if $A\mathbf{x} = \mathbf{b}$ has a solution, then the solution is obtained by translating the solution set of $A\mathbf{x} = \mathbf{0}$.

Example 1b Determine if the following nonhomogeneous system has a nontrivial solution. Then describe the solution set.

$$\begin{aligned}3x_1 + 5x_2 - 4x_3 &= 7 \\ -3x_1 - 2x_2 + 4x_3 &= -1 \\ 6x_1 + x_2 - 8x_3 &= -4\end{aligned}$$

Linear Independence and Solution Sets of Linear Systems Part III

Definition (Linear Dependence)

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is said to be *linearly dependent* if there exist scalars $c_1, c_2, \dots, c_n$, not all zero, such that:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}.$$

Definition (Linear Independence)

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is said to be *linearly independent* if the only scalars $c_1, c_2, \dots, c_n$ that satisfy:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

are $c_1 = c_2 = \dots = c_n = 0$.

Linear Independence and Solution Sets of Linear Systems Part IV

Lemma (Linear Independence and Homogeneous Systems)

The columns of a matrix A are linearly independent if and only if the homogeneous equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.

Proof.

- ($\Rightarrow$) Suppose the columns of A are linearly independent. Then, by definition, the only scalars $c_1, c_2, \dots, c_n$ that satisfy:

$$c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \cdots + c_n \mathbf{a}_n = \mathbf{0}$$

are $c_1 = c_2 = \cdots = c_n = 0$, where $\mathbf{a}_i$ are the columns of A . This implies that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.

- ($\Leftarrow$) Suppose $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$. Then there are no nontrivial combinations of the columns of A that sum to $\mathbf{0}$, implying that the columns of A are linearly independent.



Linear Independence and Solution Sets of Linear Systems Part V

Example 1: Determine if the columns of the matrix $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ are linearly independent.

Example 2: Determine if the vectors are $\begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$ are linearly independent.

Example 3: Show that the set of two vector $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other.

Theorem

Characterization of Linearly Dependent Sets An indexed set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. That is, each $\mathbf{v}_i \in S$ does not have to be a linear combination of the other vectors. Only one of them needs to satisfy the condition.

Lemma

A set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ whose elements $\mathbf{v}_i \in \mathbb{R}^n$ is linearly dependent if $p > n$.

Homework 4

Question Q1

Construct a 3×3 nonzero matrix such that the vector $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is a solution of $A\mathbf{x} = \mathbf{b}$.

Question Q2

Are the following vectors linearly dependent? Justify your answer.

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \text{ and } \mathbf{v}_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

Introduction to Linear Transformations

A **linear transformation** is a mathematical operation that maps vectors from one vector space to another (or to itself). This function can be represented as a matrix.

A linear transformation T from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be written as:

$$T(\mathbf{x}) = A\mathbf{x},$$

where: A is an $m \times n$ matrix (the transformation matrix), $\mathbf{x} \in \mathbb{R}^n$ is the input vector. $T(\mathbf{x}) \in \mathbb{R}^m$ is the transformed vector.

Note: A **transformation** is just a **function**

A transformation is linear if it has the following properties:

- All lines remain lines.
- The origin stays the same.

Example

Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$,

and define the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$.

- Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .
- Find an $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{b}$.
- Is there more than one $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{b}$?
- Determine if $\mathbf{c}$ is in the range of the transformation T . In other words, is there an $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{c}$?

Example

$$\text{Let } A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix},$$

and define the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$.

- a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .
- b. Find an $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{b}$.

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{bmatrix}$$

- c. Is there more than one $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{b}$?
- d. Determine if $\mathbf{c}$ is in the range of the transformation T . In other words, is there an $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{c}$?

Example

Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$,

and define the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$.

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- b. Find an $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{b}$.

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- c. Is there more than one $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{b}$?
- d. Determine if $\mathbf{c}$ is in the range of the transformation T . In other words, is there an $\mathbf{x} \in \mathbb{R}^2$ such that $T(\mathbf{x}) = \mathbf{c}$?

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

More Examples

Example 2: Construct a transformation matrix that projects points in $\mathbb{R}^3$ to $\mathbb{R}^2$.

Can you construct another transformation to do a similar thing?

Example 3: Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. The transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined

by $T(\mathbf{x}) = A\mathbf{x}$ is called a **shear transformation**.

What does the transformation do to a square?

Hint: apply the transformation to the points

$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, and $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$. These four points form a square of length 2 on each side.

More Examples

Example 2: Construct a transformation matrix that projects points in $\mathbb{R}^3$ to $\mathbb{R}^2$.

Can you construct another transformation to do a similar thing?

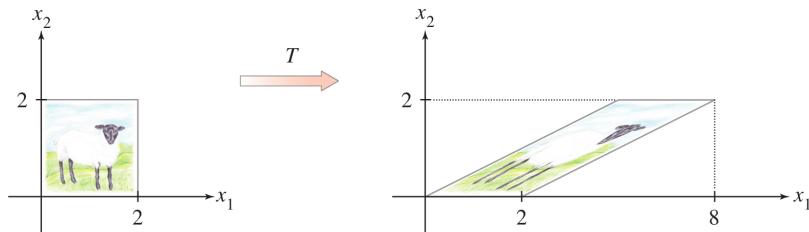
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More Examples

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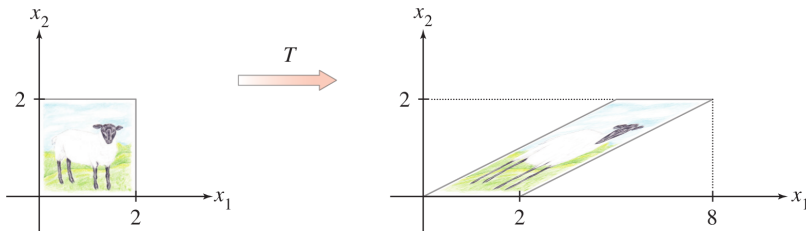
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What does the transformation do to a square?

Hint: apply the transformation to the points $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, and $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$. These four points form a square of length 2 on each side.



sheep



Definition (Linear Transformation)

A transformation T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is linear if it satisfies the following:
For any input vector $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and scalar $c \in \mathbb{R}$,

- ❶ **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.
- ❷ **Scalar Multiplication:** $T(c\mathbf{u}) = cT(\mathbf{u})$, for any scalar c .

Geometrically, linear transformations can:

- **Scale (Enlarge or Shrink)** vectors (e.g., multiplication by a scalar matrix).
- **Rotate** vectors (e.g., multiplication by a rotation matrix).
- **Reflect** vectors (e.g., across an axis or plane).
- **Shear** vectors (e.g., skew transformations).

Matrix of Linear Transformation

Examples of Transformations

1. **Scaling:**

$$A = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}, \quad T(\mathbf{x}) = A\mathbf{x}.$$

This scales all vectors by a factor k .

2. **Rotation in 2D:**

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

This rotates vectors by an angle θ .

3. **Reflection:**

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

This reflects vectors across the x -axis.

The Matrix of a Linear Transformation

Example: The columns of $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Suppose T is a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^3$ such that

$$T(\mathbf{e}_1) = \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} \quad \text{and} \quad T(\mathbf{e}_2) = \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix}.$$

With no additional information, find a formula for the image of an arbitrary $\mathbf{x} \in \mathbb{R}^2$. And hence the matrix of the transformation A .

The Matrix of a Linear Transformation

Solution:

Any $\mathbf{x} \in \mathbb{R}^2$ can be written as: $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$, where $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$,

and $x_1, x_2 \in \mathbb{R}$.

Since T is a linear transformation,

$$T(\mathbf{x}) = T(x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2) = x_1 T(\mathbf{e}_1) + x_2 T(\mathbf{e}_2).$$

Thus, the image of $\mathbf{x}$ under T is:

$$T(\mathbf{x}) = x_1 \begin{bmatrix} 5 \\ -7 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 8 \\ 0 \end{bmatrix} = \begin{bmatrix} 5x_1 \\ -7x_1 \\ 2x_1 \end{bmatrix} + \begin{bmatrix} -3x_2 \\ 8x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 5x_1 - 3x_2 \\ -7x_1 + 8x_2 \\ 2x_1 + 0 \end{bmatrix}$$

Thus, the matrix transformation, is:

$$T(\mathbf{x}) = \underbrace{\begin{bmatrix} 5 & -3 \\ -7 & 8 \\ 2 & 0 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

The Matrix of a Linear Transformation

To find the matrix A of the linear transformation T , compute the images of the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ of $\mathbb{R}^n$ under T . The i -th column of A is given by:

$$A_i = T(\mathbf{e}_i),$$

where A_i is the image of the i -th standard basis vector.

Thus, the matrix A is:

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)].$$

Theorem (Matrix Transformation)

Let T from $\mathbb{R}^n$ to $\mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that:

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

The Matrix of a Linear Transformation

Example Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by: $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 2x + 3y \\ -x + 4y \end{bmatrix}$.

Applying T to the standard basis vectors, find A .

Example Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by:

$$T\mathbf{x} = 5\mathbf{x}.$$

Example Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that rotates each point counterclockwise about the origin through an angle θ .

Find the standard matrix A of this transformation.

Example Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation that rotates each point clockwise about the origin through an angle θ .

Find the standard matrix A of this transformation.

The Matrix of a Linear Transformation

Definition (one-to-one)

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if each $\mathbf{b} \in \mathbb{R}^m$ is the image of **at most one** $\mathbf{x} \in \mathbb{R}^n$.

If $T(\mathbf{x}) = T(\mathbf{y})$ implies $\mathbf{x} = \mathbf{y}$.

That is, if any two objects map to the same image, the objects must be the same. This does not guarantee that each image has a corresponding object.

Definition (unto)

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **unto** if each $\mathbf{b} \in \mathbb{R}^m$ is the image of **at least one** $\mathbf{x} \in \mathbb{R}^n$.

That is, every image must have an object. This does not guarantee that each object has a corresponding image.

Example 1: One-to-One Transformation

A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if $T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$.

Consider the matrix: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. To check if $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one, we solve $A\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Solving for x_1 and x_2 , we rewrite the system in augmented form and row reduce:

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 4 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & -2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & 0 \end{array} \right].$$

Since $x_1 = x_2 = 0$, A is **one-to-one**.

Example 2: Onto Transformation

A transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if for every $\mathbf{b} \in \mathbb{R}^m$, there exists a solution $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$. This happens when the columns of A span $\mathbb{R}^m$.

Consider the matrix: $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. Since B is a 2×3 matrix, so it is made up of 3 2-dimensional vectors. Thus, it can only span $\mathbb{R}^2$. We row reduce it to check if it has a pivot in each row (**check theorem on page 19**):

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix}.$$

Since the matrix has a pivot in each row, the columns of B span all of $\mathbb{R}^2$, meaning B is **onto**.

The Matrix of a Linear Transformation

Example: Let T be the linear transformation whose standard matrix is

$$\begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -2 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Does T map $\mathbb{R}^4$ **onto** $\mathbb{R}^3$? Is T a **one-to-one** map?

Theorem

A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if and only if $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, and A be the standard matrix for T . Then,

- (i) T is onto if and only if the columns of A span $\mathbb{R}^m$.
- (ii) T is one-to-one if and only if the columns of A are linearly independent.

is one-to-one if and only if $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Homework 5

Question Q1: Consider a linear transformation T from $\mathbb{R}^n$ to $\mathbb{R}^m$. Using the definition of linear transformation, show that

- i) $T(c_1\mathbf{u} + c_2\mathbf{v}) = c_1T(\mathbf{u}) + c_2T(\mathbf{v})$ for any scalars $c_1, c_2 \in \mathbb{R}$.
- ii) $T(\mathbf{0}) = \mathbf{0}$.

Question Q2: Define a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}.$$

Find the images under T of $\mathbf{u} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\mathbf{v} + \mathbf{w}$. Sketch the object represented by the four coordinates, and their corresponding image under the transformation.

Matrix Operations

1. Matrix Addition and Subtraction: matrix addition and subtraction are performed element-wise. The matrices must have the same dimensions. If $A = [a_{ij}]$ and $B = [b_{ij}]$ are $m \times n$ matrices, then:

$$A + B = [a_{ij} + b_{ij}], \quad A - B = [a_{ij} - b_{ij}].$$

Example:

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}.$$

$$A + B = \begin{bmatrix} 2+1 & 4+(-2) \\ 1+3 & 3+0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}.$$

$$A - B = \begin{bmatrix} 2-1 & 4-(-2) \\ 1-3 & 3-0 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ -2 & 3 \end{bmatrix}.$$

2. Scalar Multiplication: a matrix can be multiplied by a scalar c , which means each element of the matrix is multiplied by c . If $A = [a_{ij}]$, then:

$$cA = [c \cdot a_{ij}].$$

Example:

$$A = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}, \quad c = 4.$$

$$4A = \begin{bmatrix} 4 \cdot 3 & 4 \cdot (-1) \\ 4 \cdot 0 & 4 \cdot 2 \end{bmatrix} = \begin{bmatrix} 12 & -4 \\ 0 & 8 \end{bmatrix}.$$

Matrix Operations

3. Matrix Multiplication: matrix multiplication is defined when the number of columns in the first matrix equals the number of rows in the second matrix.

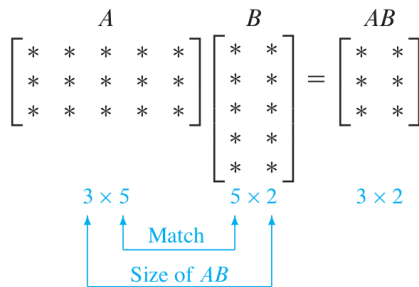
If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{jk}]$ is an $n \times p$ matrix, then the product $C = AB$ is an $m \times p$ matrix where:

$$c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}.$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}.$$

$$AB = \begin{bmatrix} (1 \cdot 2 + 2 \cdot 1) & (1 \cdot 0 + 2 \cdot (-1)) \\ (3 \cdot 2 + 4 \cdot 1) & (3 \cdot 0 + 4 \cdot (-1)) \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 10 & -4 \end{bmatrix}.$$



Matrix Operations

4. Transpose of a Matrix: the transpose of a matrix is obtained by swapping its rows with its columns. If $A = [a_{ij}]$ is an $m \times n$ matrix, then the transpose $A^T = [a_{ji}]$ is an $n \times m$ matrix.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}.$$

Theorem

Let A and B denote matrices whose sizes are appropriate for the following sums and products. The following is true:

- (i) $(A^T)^T = A$
- (ii) $(A + B)^T = A^T + B^T$
- (iii) $(AB)^T = B^T A^T$
- (iv) $(rA)^T = rA^T$

5. Determinant of a Square Matrix: the determinant is a scalar value that can be computed for square matrices.

For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is:

$$\det(A) = ad - bc.$$

Example:

$$A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}.$$

$$\det(A) = (3)(2) - (5)(1) = 6 - 5 = 1.$$

5. Determinant of a Square Matrix

For 3×3 matrix, $A = \begin{bmatrix} \textcolor{red}{a} & \textcolor{green}{b} & \textcolor{blue}{c} \\ d & e & f \\ g & h & i \end{bmatrix}$ the determinant is:

$$\det(A) = \textcolor{red}{a} \cdot \det \left(\begin{bmatrix} e & f \\ h & i \end{bmatrix} \right) - \textcolor{green}{b} \cdot \det \left(\begin{bmatrix} d & f \\ g & i \end{bmatrix} \right) + \textcolor{blue}{c} \cdot \det \left(\begin{bmatrix} d & e \\ g & h \end{bmatrix} \right)$$

Matrix Operations

6. Inverse of a Matrix The inverse of a square matrix A (if it exists) is a matrix A^{-1} such that: $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.

Formula (for 2×2 matrix): If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\det(A) \neq 0$, then:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

$$\det(A) = (2)(4) - (1)(3) = 8 - 3 = 5.$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.2 \\ -0.6 & 0.4 \end{bmatrix}.$$

7. Identity Matrix The identity matrix I_n is a square matrix with ones on the diagonal and zeros elsewhere. It acts as the multiplicative identity for matrices:

$$AI = IA = A.$$

Example:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Matrix Operations Summary

Theorem

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are define:

- (i) **Associativity:** $A(BC) = (AB)C$
- (ii) **Left Distribution:** $A(B + C) = AB + AC$
- (iii) **Right Distribution:** $(A + B)C = AC + BC$
- (iv) **Scalar Distribution:** $r(AB) = (rA)B = A(rB)$

Note: $AB \neq BA$.

Also, there are no cancellation laws. That is, if $AB = AC$, then it is not true in general that $B = C$.

Theorem

If A and B are $n \times n$ invertible matrices, then A^{-1} and B^{-1} are also invertible, and:

- i) $(A^{-1})^{-1} = A$
- ii) $(AB)^{-1} = B^{-1}A^{-1}$
- iii) **Right Distribution:** $(A + B)C = AC + BC$
- iv) $(A^{-1})^T = (A^T)^{-1}$, since A^T is also invertible.

Algorithm for Finding the Inverse of a Matrix Using Row Reductions

Algorithm Steps

- 1 Write the augmented matrix $[A \mid I_n]$.
- 2 Use row operations to transform the left-hand side A into I_n .
- 3 Ensure all pivot elements (diagonal entries) are 1, and all other elements in the column are 0.
- 4 Once the left-hand side is I_n , the right-hand side will be A^{-1} .

Conditions for Invertibility The matrix A is invertible if and only if:

- 1 A is a square matrix.
- 2 $\det(A) \neq 0$ (i.e., A is full rank).

Important Notes

- If A cannot be row reduced to I_n (e.g., a row becomes zero during the process), A is not invertible.
- The algorithm is applicable only for square matrices.

Examples

Example 1 Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$. Find A^{-1} .

Solution: Start with the augmented matrix:

$$[A \mid I_2] = \left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right].$$

Step 1: Scale the first row by $\frac{1}{2}$:

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 1 & 0 & 1 \end{array} \right].$$

Step 2: Subtract the first row from the second row: Thus, $A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \end{array} \right].$$

Step 3: Scale the second row by 2:

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & -1 & 2 \end{array} \right].$$

Step 4: Subtract $\frac{1}{2}$ of the second row from the first row:

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right]$$

Examples

Example 2 Let $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$. Find A^{-1} .

Solution: Start with the augmented matrix:

$$\left[\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 4 & 2 & 0 & 1 \end{array} \right].$$

Step 1: Scale the first row by $\frac{1}{3}$:

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 4 & 2 & 0 & 1 \end{array} \right].$$

Step 2: Subtract 4 times the first row from the second row:

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{2}{3} & -\frac{4}{3} & 1 \end{array} \right].$$

Step 3: Scale the second row by $\frac{3}{2}$:

$$\left[\begin{array}{cc|cc} 1 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 1 & -2 & \frac{3}{2} \end{array} \right].$$

Step 4: Subtract $\frac{1}{3}$ of the second row from the first row:

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -\frac{1}{2} \\ 0 & 1 & -2 & \frac{3}{2} \end{array} \right].$$

$$\text{Thus, } A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{bmatrix}.$$

Examples

Example 3: Find the inverse of $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 1 & 3 & 1 & 0 & 0 & 1 \end{array} \right].$$

Step 1: $-R_1 + R_3 \rightarrow R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{array} \right].$$

Step 2: Swap R_1 and R_3

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 & 0 \end{array} \right].$$

Step 3: $-R_2 + R_3 \rightarrow R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 & -1 \end{array} \right].$$

Step 3: $\frac{1}{2}R_3 \rightarrow R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right].$$

Continue until you obtain the inverse.

$$\text{Thus, } A^{-1} = \begin{bmatrix} \frac{5}{2} & -\frac{1}{2} & -\frac{3}{2} \\ -1 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

Homework 6

- 1 Find the inverse of $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ if it exists.
- 2 Determine whether $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is invertible. Justify your response.
- 3 Verify that the inverse you found for $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ satisfies $AA^{-1} = I$.
- 4 Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$.
- 5 Verify that the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 2 \\ 3 & -2 & 1 \end{bmatrix}$ is invertible and find its inverse.
- 6 Show that the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is not invertible.

Theorem (Invertible Matrix Theorem)

Let A be an $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either **all TRUE** or **all FALSE**.

- (a) A is an invertible matrix. A^{-1} is also invertible.
- (b) A is row equivalent to an $n \times n$ identity matrix.
- (c) A has a pivot in each row after row reduction.
- (d) The matrix equation, $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b} \in \mathbb{R}^n$.
- (e) The columns of A span $\mathbb{R}^n$.
- (f) The columns of A are linearly independent.
- (g) The homogeneous equation, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (h) The linear transformation $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one
- (i) The linear transformation $T(\mathbf{x}) = A\mathbf{x}$ is onto.

Characterization of Invertible Matrices

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation such that $T(\mathbf{x}) = A\mathbf{x}$, where A is the transformation matrix. Then T is invertible if and only if A is an invertible matrix.

And the inverse of the transformation $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is $T^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}$

Homework 7

- 1 Is $A = \begin{bmatrix} 0 & 2 & -2 & 3 \\ 0 & 0 & 0 & 5 \\ 1 & -4 & 8 & 1 \end{bmatrix}$ one-to-one? Explain the reason for your answer.
- 2 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by: $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 2x + 3y \\ x + 4y \end{bmatrix}$. Write out T in the form, $T(\mathbf{x}) = A\mathbf{x}$.
Is T invertible? If so, find a formula for T^{-1} .

Definition

A **subspace** of $\mathbb{R}^n$ is any set $S \subseteq \mathbb{R}^n$ that has 3 properties:

- i) The zero vector is in S .
- ii) For each $\mathbf{u}, \mathbf{v} \in S$, $\mathbf{u} + \mathbf{v} \in S$.
- iii) For each $\mathbf{u} \in S$, and scalar $c \in \mathbb{R}$, $c\mathbf{u} \in S$.

Example: If $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^n$, and $S = \text{Span}(\{\mathbf{u}_1, \mathbf{u}_2\})$. Then S is a subspace of $\mathbb{R}^n$.

Definition

A **column space** of matrix A is the set, $ColA = Span(\{\text{Columns of } A\})$. That is, the set of all linear combinations of the columns of A .

Example: Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ 3 \\ -4 \end{bmatrix}$. Determine whether $\mathbf{b}$ is in the column space of A .

Definition

A **null space** of matrix A is the set, $Nul A$ - of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Theorem

A **null space** of and $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of the homogeneous system $A\mathbf{x} = \mathbf{0}$ is also a subspace of $\mathbb{R}^n$.

Definition

A **basis** of a subspace S of $\mathbb{R}^n$ is a linearly independent set in S that spans S .

Note: The number of pivots tells you the number of basis elements the **null space** and **column space**.

$$\# \text{ of basis elements in null space} + \# \text{ of basis elements in column space} = \# \text{ of columns}$$

Theorem

A ***pivot columns*** of a matrix A form a basis for the columns space of A .

Definition

A **basis** of a subspace S of $\mathbb{R}^n$ is a linearly independent set in S that spans S .

Note: The number of pivots tells you the number of basis elements the **null space** and **column space**.

$$\# \text{ of basis elements in null space} + \# \text{ of basis elements in column space} = \# \text{ of columns}$$

Example 1: Find the basis for the null space, and the basis of the column space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Definition

A **basis** of a subspace S of $\mathbb{R}^n$ is a linearly independent set in S that spans S .

Note: The number of pivots tells you the number of basis elements the **null space** and **column space**.

$$\# \text{ of basis elements in null space} + \# \text{ of basis elements in column space} = \# \text{ of columns}$$

Example 2: Find the basis for the null space, and the basis of the column space of the matrix

$$A = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Question 1: Find the basis for the null space, and the basis of the column space of the matrix

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

Dimension of a Subspace

Definition

The **dimension** of a nonzero subspace S , denoted by $\dim S$ is the number of vectors in a basis for the subspace S .

The dimension of the zero subspace $\{0\}$ is defined to be zero.

Definition

The **rank** of a matrix A , denoted by $\text{rank } A$ is the dimension of the column space of A .

The dimension of the zero subspace $\{0\}$ is defined to be zero.

Example 1: Determine the rank and $\dim(\text{Nul}A)$ of the matrix below

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Example 2: determine the rank and $\dim(\text{Nul}A)$ of the matrix below

$$B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Theorem (Invertible Matrix Theorem Continued)

Let A be an $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either **all TRUE** or **all FALSE**.

- (a) A is an invertible matrix. A^{-1} is also invertible.
- (c) A has a pivot in each row after row reduction.
- (j) Let A be and $n \times n$ matrix .
- (k) The columns of A form a basis of $\mathbb{R}^n$.
- (l) The column space of A , $\text{Col}A = \mathbb{R}^n$.
- (m) The dimension of the column space of A , $\dim \text{Col}A = n$.
- (n) $\text{rank}A = n$.
- (o) The null space of A , $\text{Nul}A = \{\mathbf{0}\}$.
- (p) The dimension of the null space of A , $\dim \text{Nul}A = 0$.

Introduction to Determinants

Intuition: Geometrically, the **determinant of a matrix A measures the scaling factor** after applying the matrix transformation $T(\mathbf{x}) = A\mathbf{x}$.

In a 2×2 matrix, the scaling factor measures how much bigger/smaller the area of the unit square gets after the transformation has been applied. For a 3×3 matrix, how much bigger volume of the unit cube gets after the transformation has been applied.

If $\det(A) = 0$, the transformation collapses space into a lower dimension, implying that A is non-invertible (singular/ does not have an inverse).

For example:

- If a 2×2 matrix has a determinant of 5, it scales areas by a factor of 5.
- If a 3×3 matrix has a determinant of -2, it scales volumes by a factor of 2 and reverses orientation (flips it like a page).

Definition (Determinant)

The determinant of a square matrix A of order $n \times n$ is a scalar value that provides important information about the matrix, such as whether it is invertible and how it scales volumes in linear transformations. The determinant of a matrix A is denoted as $\det(A)$ or $|A|$.

For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ its determinant is given by: } \det(A) = ad - bc.$$

For a 3×3 matrix:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, \quad \det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

Introduction to Determinants

Example 1

Given the matrix:

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix},$$

its determinant is:

$$\det(A) = (3)(4) - (2)(1) = 12 - 2 = 10.$$

Example 2

Consider the matrix:

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

its determinant is:

$$\det(B) = (1)(4) - (2)(2) = 4 - 4 = 0.$$

Since the determinant is zero, B is singular and non-invertible.

Definition (Generalized Definition of Determinant)

The **determinant** of a square matrix $A = [a_{ij}]$ of order $n \times n$ is the sum of n terms of the form $\pm a \det(A_{ij})$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A (**not entirely true**). **Matrix A_{ij} is obtained by deleting the i th row and the j th column.**

$$\begin{aligned}\det(A) \text{ or } |A| &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}) \\ &= a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + a_{13} \det(A_{13}) - \cdots + (-1)^{1+n} a_{1n} \det(A_{1n})\end{aligned}$$

Example: Compute the determinant of $A = \begin{bmatrix} 2 & 4 & -1 \\ 0 & -2 & 0 \\ 1 & 5 & 0 \end{bmatrix}$

Definition (Cofactor Definition of Determinant)

The **cofactor** of a matrix A is the number C_{ij} given by $C_{ij} = (-1)^{i+j} \det(A_{ij})$.

The **determinant** of a square matrix $A = [a_{ij}]$ of order $n \times n$ can be computed using a **cofactor expansion** across the first row of A (not entirely true).

$$\det(A) \text{ or } |A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + \cdots + a_{1n}C_{1n}$$

Example: Compute the determinants of $A = \begin{bmatrix} 2 & 4 & -1 \\ 0 & -2 & 0 \\ 1 & 5 & 0 \end{bmatrix}$, and $B = \begin{bmatrix} 2 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$

Introduction to Determinants

Recall: when we row reduce a matrix, it does not change the matrix.
Hence this can make the computation of the determinant easier!

Theorem (Determinant Theorem)

*If A is a **triangular matrix**, then $\det A$ is a product of the entries on the main diagonal of A .*

Theorem (Row Operations)

Let A a square matrix.

- Ⓐ If a multiple of one row of A is added to another row to produce a matrix B , then $\det A = \det B$.*
- Ⓑ If two rows of A are interchanged to produce a matrix B , then $\det A = -\det B$.*
- Ⓒ If a row of A is multiplied by k to produce a matrix B , then $\det B = k \cdot \det A$.*

Theorem (Properties of Determinants)

- a** **Determinant of Identity:** The determinant of the identity matrix I_n is 1, i.e., $\det(I_n) = 1$.
- b** **Row or Column Swap:** Swapping two rows or columns of a matrix A to produce B negates its determinant:

$$\det(B) = -\det(A),$$

- c** **Multiplication by a Scalar:** Multiplying a row or a column by a scalar k multiplies the determinant by k : Therefore multiplying all n rows of a matrix by k ,

$$\det(kA) = k^n \det(A).$$

- d** **Triangular Matrices:** The determinant of a triangular (upper or lower) matrix is the product of its diagonal elements:

$$\det(A) = a_{11}a_{22} \dots a_{nn}.$$

Theorem (Properties of Determinants)

- (e) **Determinant of a Product:** *The determinant of the product of two matrices is the product of their determinants:*

$$\det(AB) = \det(A) \cdot \det(B).$$

- (f) **Effect of Row Operations:** *Adding a multiple of one row to another does not change the determinant.*

- (g) **Invertibility:** *A square matrix A is invertible if and only if $\det(A) \neq 0$.*

- (h) **Transpose Property:** *The determinant of a matrix is equal to the determinant of its transpose:*

$$\det(A^T) = \det(A).$$

Cramer's Rule, Volume, and Linear Transformations

Cramer's Rule Cramer's rule is needed in a variety of theoretical calculations. It can be used to study how the solution of $A\mathbf{x} = \mathbf{b}$ is affected by changes in the entries of $\mathbf{b}$. However, the formula is inefficient for hand calculations, except for 2×2 or 3×3 matrices.

Cramer's Rule provides an explicit formula for solving a system of linear equations using determinants. Given a system of n linear equations in n variables represented in matrix form as:

$$A\mathbf{x} = \mathbf{b},$$

where A is an $n \times n$ invertible matrix, the solution for each variable x_i is given by:

$$x_i = \frac{\det(A_i)}{\det(A)},$$

where A_i is the matrix obtained by replacing the i -th column of A with the column vector $\mathbf{b}$.

Cramer's Rule, Volume, and Linear Transformations

Example 1: Consider the system,

$$2x_1 + 3x_2 = 5,$$

$$4x_1 + x_2 = 6.$$

The coefficient matrix is:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix},$$

with determinant:

$$\det(A) = (2)(1) - (3)(4) = 2 - 12 = -10.$$

Replacing the first column with $\mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ gives $A_1 = \begin{bmatrix} 5 & 3 \\ 6 & 1 \end{bmatrix}$, with

$\det(A_1) = (5)(1) - (3)(6) = 5 - 18 = -13$. Thus,

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{-13}{-10} = 1.3.$$

Repeating for A_2 yields $x_2 = 0.8$.

Definition (Cofactor Formula for Inverse A^{-1})

Let A be an invertible $n \times n$ matrix. Then

$$\begin{aligned} A^{-1} &= \frac{1}{\det A} \cdot \operatorname{adj} A \\ &= \frac{1}{\det A} \cdot \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} \end{aligned}$$

$C_{ij} = (-1)^{i+j} \det(A_{ij})$. **Matrix A_{ij} is obtained by deleting the i th row and the j th column.**

Theorem (Determinant-Area)

If A is a 2×2 matrix,

the area of the parallelogram determined by the columns of $A = |\det A|$.

Theorem (Determinant-Area)

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation determined by 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\}.$$

Example:

Let S be the parallelogram determined by $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$.

Compute the area of the image of S under the mapping $\mathbf{x} \rightarrow A\mathbf{x}$.

Definition (Vector Space)

A nonempty set V , is a **vector space** if it satisfies the following:

- a) For every $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} \in V$.
- b) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- c) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$, where $\mathbf{w} \in V$.
- d) There is a **zero** vector $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
- e) For every $\mathbf{u} \in V$, there is a vector $-\mathbf{u} \in V$ such that $\mathbf{u} + -\mathbf{u} = \mathbf{0}$.
- f) For every $c \in \mathbb{R}$, $c\mathbf{u} \in V$.
- g) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
- h) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.
- i) $c(d\mathbf{u}) = (cd)\mathbf{u}$.
- j) $1\mathbf{u} = \mathbf{u}$.

Vector Spaces and Subspaces

Example1: For $n \geq 0$, the set $\mathbb{P}_n$ of polynomials of degree at most n consists of all polynomials of the form

$$\mathbf{P}(t) = a_0 + a_1t + a_2t^2 + \cdots + a_nt^n$$

each form a vector space.

Example2: The space of n dimensional real number vectors, $\mathbb{R}^n$ is a vector space.

Definition (Subspace)

A **subspace** of $\mathbb{R}^n$ is any set $S \subseteq \mathbb{R}^n$ that has 3 properties:

- i) The zero vector is in S .
- ii) For each $\mathbf{u}, \mathbf{v} \in S$, $\mathbf{u} + \mathbf{v} \in S$.
- iii) For each $\mathbf{u} \in S$, and scalar $c \in \mathbb{R}$, $c\mathbf{u} \in S$.

Vector Spaces and Subspaces

Example1: The set consisting of only the zero vector in a vector space V is a subspace of V , called the **zero subspace** and written as $\{\mathbf{0}\}$.

Example2: The set of all 3D vectors whose first component is $\mathbf{0}$ is a subspace of $\mathbb{R}^3$.

Example3: The $\text{Span}(\{\mathbf{v}_1, \mathbf{v}_2\})$, where $\mathbf{v}_1, \mathbf{v}_2 \in V$ of an arbitrary vector space is a subspace of V .

Theorem (Subspace Theorem)

If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are vectors in a vector space V , then $\text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_p\})$ is a subspace of V .

Example: Show that H is a subspace of $\mathbb{R}^3$.

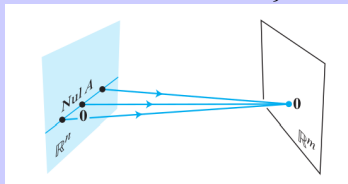
$$H = \left\{ s \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + t \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} : \text{for } s, t \in \mathbb{R} \right\}$$

Null Spaces and Column/Range Spaces

Definition (Null Space)

A **null space** of matrix A is the set, $Nul A$ - of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$$Nul A = \{\mathbf{x} : \mathbf{x} \in \mathbb{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\}$$



Theorem

A **null space** of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of the

Null Spaces and Column/Range Spaces

Definition (Column Space)

A **column space** or **range space** of matrix A is the set, $ColA = Span(\{\text{Columns of } A\})$. That is, the set of all linear combinations of the columns of A .

Theorem

The **column space** of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$. Equivalently, the set of all $\mathbf{b} \in \mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{b}$ is the column space or range space (the range of the linear transformation $A\mathbf{x}$).

Null Spaces and Column/Range Spaces

Example 1: Find the null space, $(\text{Nul}A)$ of the matrix below

$$A = \begin{bmatrix} 1 & 2 & -4 & 3 & -1 \\ -3 & -2 & 1 & 0 & -1 \\ 2 & -5 & 1 & 8 & -3 \end{bmatrix}$$

Example 2: Find the null space, $(\text{Nul}A)$ of the matrix below

$$B = \begin{bmatrix} 1 & -3 & 5 & 0 \\ 1 & 2 & -1 & 0 \\ 0 & 7 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

Definition (Linear Transformation)

A transformation T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is linear if it satisfies the following:
For any input vector $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and scalar $c \in \mathbb{R}$,

- i) **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.
- ii) **Scalar Multiplication:** $T(c\mathbf{u}) = cT(\mathbf{u})$, for any scalar c .

Null Spaces and Column/Range Spaces

Example 1: Find the column/range space, $(\text{Col}A)$ of the matrix below

$$A = \begin{bmatrix} 1 & 2 & -4 & 3 & -1 \\ -3 & -2 & 1 & 0 & -1 \\ 2 & -5 & 1 & 8 & -3 \end{bmatrix}$$

Example 2: Find the column/range space, $(\text{Col}A)$ of the matrix below

$$B = \begin{bmatrix} 1 & -3 & 5 & 0 \\ 1 & 2 & -1 & 0 \\ 0 & 7 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

Contrast between Range/Column space and Null space/Kernel of an $m \times n$ matrix, A

- 1 $\text{Nul}A$ is a subspace of $\mathbb{R}^n$
 - 2 $\text{Nul}A$ is the solution to $A\mathbf{x} = \mathbf{0}$. You have to find $\mathbf{x}$ in order to find the null space.
 - 3 There is no obvious relation between $\text{Nul}A$ and the columns or row of A .
 - 4 Any vector $\mathbf{v}$ is in the null space of A if and only if $A\mathbf{v} = \mathbf{0}$.
 - 5 $\text{Nul } A = \{\mathbf{0}\}$ if and only if the only solution to $A\mathbf{x} = \mathbf{0}$ is the zero vector.
- 1 $\text{Col}A$ is a subspace of $\mathbb{R}^m$
 - 2 $\text{Col}A$ is a linear combination of some of the columns of A . **Select the columns that have pivots after row reduction.**
 - 3 The column vectors of A are in $\text{Col}A$.
 - 4 Any vector $\mathbf{b}$ is in the columns space of A if and only if $A\mathbf{x} = \mathbf{b}$ is a consistent system. (You need to row reduce).
 - 5 $\text{Col } A = \mathbb{R}^m$ if and only if $A\mathbf{x} = \mathbf{b}$ has a pivot in each row.

Definition (Linear Independence)

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is said to be **linearly independent** if the only scalars $c_1, c_2, \dots, c_n$ that satisfy:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

are $c_1 = c_2 = \dots = c_n = 0$.

In other words, the solution to the homogeneous solution $A\mathbf{x} = \mathbf{0}$ is the trivial solution.

Definition (Basis)

Let H be a subspace of a vector space V . A set $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p\}$ is a basis in V for H if

- (i) $\mathcal{B}$ is a linearly independent set
- (ii) $\text{Span}(\mathcal{B}) = H$

Theorem (The Basis Theorem)

Let V be an n -dimensional vector space V . Any linearly independent set $\mathcal{L} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V .

That is, any linearly independent set with exactly n elements is a basis.

Examples

Example 1: Find the basis of $\text{Col}A$ and $\text{Nul}A$ of the matrix below

$$A = \begin{bmatrix} 1 & 2 & -4 & 3 & -1 \\ -3 & -2 & 1 & 0 & -1 \\ 2 & -5 & 1 & 8 & -3 \\ -1 & 0 & 3 & 1 & 0 \end{bmatrix}$$

Example 2: Find the basis of $\text{Col}A$ and $\text{Nul}A$ of the matrix below

$$B = \begin{bmatrix} 1 & -3 & 5 & 0 \\ 1 & 2 & -1 & 0 \\ 0 & 7 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

Example 3: Are $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 7 \end{bmatrix} \right\}$ and $\mathcal{D} = \left\{ \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -3 \end{bmatrix} \right\}$ bases?

Theorem (Unique Representation)

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ be a basis for a vector space V . Then for each $\mathbf{x} \in V$, there exists a unique set of scalars $c_1, c_2, \dots, c_p$ such that $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_p\mathbf{b}_p = \mathbf{x}$.

The **basis** imposes a coordinate system for that vector space.

Definition (Basis)

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ be a basis for a vector space V . The coordinates of $\mathbf{x}$ relative to the basis $\mathcal{B}$ (or the $\mathcal{B}$ -coordinates of $\mathbf{x}$) are a unique set of scalars $c_1, c_2, \dots, c_p$ such that $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_p\mathbf{b}_p = \mathbf{x}$.

$$\begin{bmatrix} c_1 \end{bmatrix}$$

If $c_1, \dots, c_p$ are the $\mathcal{B}$ -coordinates of $\mathbf{x} \in \mathbb{D}^n$, $\begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$ is the $\mathcal{B}$ -coordinate vector of

Examples

Example 1: Let a basis $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \end{bmatrix} \right\}$. Suppose that $\mathbf{x} \in \mathbb{R}^2$ has the coordinate vector $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$. Find $\mathbf{x}$.

Example 2: Find the coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ of $\mathbf{x}$ from example 1 relative to the **standard basis** $\mathcal{D} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$.

Example 3: Let a basis $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 5 \end{bmatrix} \right\}$. $H = \text{Span}(\mathcal{B})$ is a subspace. Determine whether $\mathbf{x} = \begin{bmatrix} -3 \\ 3 \\ 15 \end{bmatrix}$ is in the subspace H , and find its coordinate vector $[\mathbf{x}]_{\mathcal{B}}$.

Change of Basis

A **change of basis** allows us to represent vectors and linear transformations with respect to any coordinate systems of our choice.

Definition (Change of Basis)

Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}$ be two different bases of an n -dimensional vector space V .

Let $[\mathbf{v}]_{\mathcal{B}}$ and $[\mathbf{v}]_{\mathcal{C}}$ represent the $\mathcal{B}$ - coordinate and $\mathcal{C}$ - coordinate of the vector $\mathbf{v}$ in bases $\mathcal{B}$ and $\mathcal{C}$ respectively, then its coordinate change form $[\mathbf{v}]_{\mathcal{B}}$ and $[\mathbf{v}]_{\mathcal{C}}$ is defined as:

$$[\mathbf{v}]_{\mathcal{C}} = P_{\mathcal{B} \rightarrow \mathcal{C}} [\mathbf{v}]_{\mathcal{B}}$$

where $P_{\mathcal{B} \rightarrow \mathcal{C}}$ is the **change of basis matrix** from basis $\mathcal{B}$ to basis $\mathcal{C}$

Finding the Change of Basis Matrix

The matrix $P_{\mathcal{B} \rightarrow \mathcal{C}}$ is found by expressing each vector in $\mathcal{B}$ in terms of the basis $\mathcal{C}$.

$$\text{If } \mathbf{b}_i = a_{1i}\mathbf{c}_1 + a_{2i}\mathbf{c}_2 + \cdots + a_{ni}\mathbf{c}_n, \quad \text{then } [\mathbf{b}_i]_{\mathcal{C}} = \begin{bmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{bmatrix}$$

$$\text{and } P_{\mathcal{B} \rightarrow \mathcal{C}} \text{ has columns } P_{\mathcal{B} \rightarrow \mathcal{C}} = \begin{bmatrix} | & | & & | \\ [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \\ | & | & & | \end{bmatrix}.$$

Alternatively, if we know $P_{\mathcal{C} \rightarrow \mathcal{B}}$, then

$$P_{\mathcal{B} \rightarrow \mathcal{C}} = P_{\mathcal{C} \rightarrow \mathcal{B}}^{-1}.$$

Examples

Example 1: Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$ be a basis for $\mathbb{R}^2$. Express the vector $\mathbf{v} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$ in the $\mathcal{B}$ -basis.

Solution: Find scalars c_1, c_2 such that

$$\mathbf{v} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

This gives the system:

$$\begin{cases} c_1 + 2c_2 = 5, \\ c_1 + 3c_2 = 7. \end{cases} \Rightarrow \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & | & 5 \\ 1 & 3 & | & 7 \end{bmatrix}$$

Solving, we get $c_1 = 1$, $c_2 = 2$, so $[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{\mathcal{B}}$.

Examples

Example 2: Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$, $\mathcal{C} = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix} \right\}$ find the **change of basis matrix** $P_{\mathcal{B} \rightarrow \mathcal{C}}$.

Solution: Express each vector in $\mathcal{B}$ in terms of $\mathcal{C}$:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} = a_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 4 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \Rightarrow \left[\begin{array}{cc|c} 2 & 4 & 1 \\ 1 & 3 & 2 \end{array} \right], \text{ solve for } a_1 \text{ and } a_2$$

$$\begin{bmatrix} 3 \\ 5 \end{bmatrix} = b_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + b_2 \begin{bmatrix} 4 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \Rightarrow \left[\begin{array}{cc|c} 2 & 4 & 3 \\ 1 & 3 & 5 \end{array} \right], \text{ solve for } b_1 \text{ and } b_2.$$

Solving both systems, we get $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -\frac{5}{2} \\ \frac{3}{2} \end{bmatrix}$ and $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} -\frac{11}{2} \\ \frac{7}{2} \end{bmatrix}$. So, $P_{\mathcal{B} \rightarrow \mathcal{C}} = \begin{bmatrix} -\frac{5}{2} & -\frac{11}{2} \\ \frac{3}{2} & \frac{7}{2} \end{bmatrix}$.

Alternatively, solve

$$\left[\begin{array}{cc|cc} 2 & 4 & 1 & 3 \\ 1 & 3 & 2 & 5 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & -\frac{5}{2} & -\frac{11}{2} \\ 0 & 1 & \frac{3}{2} & \frac{7}{2} \end{array} \right]$$

Try your hands on the following problems:

- 1 Given $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$, describe the steps to compute the change of basis matrix $P_{\mathcal{B} \rightarrow \mathcal{C}}$.
- 2 If the transformation matrix of T in the standard basis is:

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix},$$

what is its representation in a new basis $\mathcal{B}$ with basis vectors $\mathbf{b}_1 = (3, 1)$ and $\mathbf{b}_2 = (2, -1)$?

Hint: to get the representation of T in the basis $\mathcal{B}$, you need to *transform your vector from the $\mathcal{B}$ -coordinate to the standard coordinate system $\mathcal{E}$* , *then apply the transformation*, and *then return the vector back to the $\mathcal{B}$ -coordinate system*. Now multiply all 3 in that order to get the new transformation matrix in the $\mathcal{B}$ -coordinate system.

- 3 Let $\mathcal{B} = \{(2, 1), (1, 3)\}$ and $\mathcal{C} = \{(3, 2), (4, 1)\}$. Find the matrix $P_{\mathcal{B} \rightarrow \mathcal{C}}$.

Homework 10

Question 1: Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be the basis: $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{b}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$. Express the vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ in the $\mathcal{B}$ -basis.

Question 2: Let: $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} \right\}$, $\mathcal{C} = \left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$. Find the change of basis matrix $P_{\mathcal{B} \rightarrow \mathcal{C}}$.

Question 3: If the transformation matrix of T in the basis $\mathcal{H}$ with basis vectors $\mathbf{h}_1 = (3, 1)$ and $\mathbf{h}_2 = (0, 1)$ is:

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix},$$

what is its representation in a new basis $\mathcal{B}$ with basis vectors $\mathbf{b}_1 = (3, 1)$ and $\mathbf{b}_2 = (2, -1)$?

Eigenvectors and Eigenvalues

Definition (Eigenvectors)

Given a square matrix A , a nonzero vector $\mathbf{v}$ is an eigenvector of A if there exists a scalar λ such that:

$$A\mathbf{v} = \lambda\mathbf{v}$$

where λ is called the **eigenvalue** corresponding to $\mathbf{v}$.

Example 1: Given the matrix $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, are $\mathbf{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ eigenvectors of A ?

Example 2: Given $B = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$, are $\mathbf{u} = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ eigenvectors of B ?

Computing Eigenvalues and Eigenvectors

To find the eigenvalues of a matrix A , solve the characteristic equation:

$$\det(A - \lambda I) = 0 \quad \text{where } I \text{ is the identity matrix of the same dimension as } A.$$

Once an eigenvalue λ is found, we obtain it's corresponding eigenvector $\mathbf{v}$ by solving:

$$(A - \lambda I)\mathbf{v} = 0$$

for each eigenvalue λ .

Computing Eigenvalues and Eigenvectors

Example 1: Find the eigenvalues and eigenvectors for the matrix: $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$

❶ **Compute the Characteristic Equation** $\det \left(\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$

$$\det \left(\begin{bmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} \right) = 0 \Rightarrow (4 - \lambda)(3 - \lambda) - (2)(1) = 0$$

$$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0 \Rightarrow \lambda^2 - 7\lambda + 10 = 0 \Rightarrow \lambda = 5, 2$$

❷ **Finding the Eigenvectors**

for $\lambda = 5$,

$$(A - 5I)\mathbf{v} = 0 \Rightarrow \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

For $\lambda = 2$,

$$(A - 2I)\mathbf{v} = 0 \Rightarrow \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Computing Eigenvalues and Eigenvectors

Example 2: Find its eigenvalues and eigenvectors for the matrix $B = \begin{bmatrix} 6 & -1 & 0 \\ -1 & 6 & -1 \\ 0 & -1 & 6 \end{bmatrix}$

Compute the Characteristic Equation $\det(B - \lambda I) = \begin{vmatrix} 6 - \lambda & -1 & 0 \\ -1 & 6 - \lambda & -1 \\ 0 & -1 & 6 - \lambda \end{vmatrix} = 0$

Ans: $\lambda_1 = 6, \mathbf{v}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}; \quad \lambda_2 = 6 - \sqrt{2}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}; \quad \lambda_3 = 6 + \sqrt{2}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$

Example 3: Find its eigenvalues and eigenvectors of $D = \begin{bmatrix} 6 & -1 & 0 \\ -1 & 6 & -1 \\ 0 & 0 & 6 \end{bmatrix}$

Ans: $\lambda_1 = 5, \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}; \quad \lambda_2 = 6, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}; \quad \lambda_3 = 7, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix};$

Computing Eigenvalues and Eigenvectors

Additional Practice Problems (Homework)

- 1 Find the eigenvalues and eigenvectors of

$$C = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

- 2 Compute the eigenvalues and eigenvectors of the 3×3 matrix:

$$D = \begin{bmatrix} -26 & -33 & -2 \\ 31 & 42 & 23 \\ -11 & -15 & -4 \end{bmatrix}$$

- 3 Prove that the eigenvalues of an upper triangular matrix correspond to the values on the main diagonal.
- 4 **Extra-if you want to try:** Show that if $\mathbf{v}_1, \dots, \mathbf{v}_n$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_n$ of an $n \times n$ matrix A , then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly independent.

More on Eigenvectors and Eigenvalues

Theorem

*Let A be an $n \times n$ matrix. Then A is invertible if and only if $\lambda = 0$ is **not** an eigenvalue of A .*

Definition (Similarity)

An $n \times n$ matrix A , is **similar** to an $n \times n$ matrix B if there exists an invertible matrix P such that $P^{-1}AP = B$.

If A and B are **similar**, then they have the same characteristic polynomial, and hence the same eigenvalues (with the same multiplicities.)

Diagonalization

An $n \times n$ matrix D is **diagonal** if all its entries off the main diagonal are zero.

Diagonalization is the process of converting a square matrix A into a diagonal matrix D by means of a *similarity transformation*.

Definition (Diagonalizability)

A square matrix A is **diagonalizable** if there exists an invertible matrix P and a diagonal matrix D such that:

$$A = PDP^{-1}$$

where D is a diagonal matrix with eigenvalues of A on the diagonal. The columns of P are the eigenvectors of A .

In other words, A is similar to D .

Theorem (Conditions for Diagonalizability)

An $n \times n$ matrix, A is diagonalizable if and only if:

- ❶ *It has n linearly independent eigenvectors.*
- ❷ *The algebraic multiplicity of each eigenvalue equals its geometric multiplicity (If an eigenvalue has a multiplicity of 2, then that eigenvalue has 2 corresponding eigenvectors).*
- ❸ *It has eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, and the matrix of eigenvectors $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$ is invertible.*

Corollary (Steps to Diagonalize a Matrix)

To diagonalize a diagonalizable matrix, A :

- ① **Find the eigenvalues:** Solve $\det(A - \lambda I) = 0$.
- ② **Find the eigenvectors:** Solve $(A - \lambda I)x = 0$ for each eigenvalue λ .
- ③ **Check independence:** Ensure there are n linearly independent eigenvectors.
- ④ **Form P and D :**
 - P consists of the eigenvectors as column
 - D is a diagonal matrix with eigenvalues on the diagonal.
- ⑤ Compute $P^{-1}AP$ to confirm D to **verify**.

Useful things to note:

- Symmetric Matrices are always diagonalizable, and their eigenvectors are orthogonal (we will talk about this in the next module).

Theorem (The Diagonalization Theorem)

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

NOTE: this **does not** imply that, if a matrix **does not have** n **distinct eigenvalues**, then it is **not diagonalizable**.

Check **Conditions for Diagonalizability Theorem**

Example 1:

Diagonalize the matrix $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$ if possible. And find a formula for A^k , given that $A = PDP^{-1}$.

Examples

Example 2:

Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{bmatrix}.$$

Solution

The eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = 1$, $\lambda_3 = -1$. So by the diagonalization theorem, A is diagonalizable since it has 3 distinct eigenvalues.

It has corresponding eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ respectively.

$$\text{So, } P = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix} \text{ and } D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix},$$

where $A = PDP^{-1}$.

Example 3:

Diagonalize the matrix, $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. if possible.

Solution

The eigenvalues of A are $\lambda_1 = 2$, $\lambda_{2,3} = -1$. So the **diagonalization theorem** cannot guarantee that A is diagonalizable since it does not have 3 distinct eigenvalues.

It has corresponding eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$,

Continue the solution to see whether A is diagonalizable. If $\lambda_{2,3}$ yields two eigenvectors, then it is. And you can go ahead to find the corresponding P and D .

Examples

Example 4:

Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}.$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

Example 5:

Determine whether $A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$ is diagonalizable.

Theorem

Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \lambda_2, \dots, \lambda_p$.

- a. For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k .
- b. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n , and this happens if and only if
 - the characteristic polynomial factors completely into linear factors
 - the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k
- c. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k for each k , then the total collection of vectors in the set $\mathcal{B}_1, \dots, \mathcal{B}_p$ forms an eigenvector basis for $\mathbb{R}^n$

Examples

Diagonalize the following matrices if possible

a) $A = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$

b) $A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$

c) $A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$

d) $A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$

e) $A = \begin{bmatrix} -6 & 4 & 0 & 9 \\ -3 & 0 & 1 & 6 \\ -1 & -2 & 1 & 0 \\ -4 & 4 & 0 & 7 \end{bmatrix}$

Homework 11

- 1 Determine whether the following are diagonalizable: If so, diagonalize them.

a)

$$A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

b)

$$A_2 = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}.$$

c)

$$A_3 = \begin{bmatrix} 5 & -8 & 1 \\ 0 & -3 & 7 \\ 0 & 0 & -2 \end{bmatrix}.$$

- 2 Determine whether $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are similar.

Complex Eigenvalues

- 1 Find the eigenvalues and corresponding eigenvectors of $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
- 2 Find the eigenvalues and corresponding eigenvectors

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{3}{5} \\ \frac{3}{4} & \frac{11}{10} \end{bmatrix}$$

Inner Product, Length, and Orthogonality

The **inner product** provides *a way to measure the similarity between two vectors*.

Definition

Inner Product For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the inner product (or dot product) is:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

Theorem

Properties of Inner Product For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and scalar c :

- a) $\mathbf{u} \cdot \mathbf{v} \in \mathbb{R}$.
- b) **Commutativity:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- c) **Distributivity:** $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
- d) **Scalar Multiplication:** $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v})$
- e) **Non-Negativity:** $\mathbf{u} \cdot \mathbf{u} \geq 0$, with equality only if $\mathbf{u} = \mathbf{0}$.

Definition

Length (Norm) of a Vector The **length** (or **norm**) of a vector $\mathbf{v}$ in $\mathbb{R}^n$ is given by:

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

The **length** or **distance** between two vectors $\mathbf{u}$ and $\mathbf{v}$ is the **length** of difference vector. That is,

$$\|\mathbf{u} - \mathbf{v}\|$$

Inner Product, Length, and Orthogonality

Example 1: Find the distance between the vectors $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$

Example 2: Find the length of the vector $\mathbf{v} = \begin{bmatrix} -3 \\ 1 \\ 6 \end{bmatrix}$.

Example 3: Find the inner product between the vectors $\mathbf{u}_1 = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 7 \\ 0 \\ -2 \end{bmatrix}$

Definition

Orthogonal Vectors Two vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal** (perpendicular) if their inner product is zero:

$$\mathbf{u} \cdot \mathbf{v} = 0$$

Example 4: Prove that $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ if the vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal**.

Example 5: Prove that $(\|\mathbf{u}\| + \|\mathbf{v}\|)^2 \geq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ for all the vectors $\mathbf{u}$ and $\mathbf{v}$.

Example 6: Prove that $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$ for all the vectors $\mathbf{v}$.

Definition

Relations between Inner Product and Angles The inner product of two vectors $\mathbf{u}, \mathbf{v}$ can also be expressed in terms of their direction as:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between the two vectors.

Orthogonal Sets

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is **orthogonal** if each pair of distinct vectors within the set $\mathbf{v}_i$ and $\mathbf{v}_j$ are **orthogonal**. That is, $\mathbf{v}_i \cdot \mathbf{v}_j = 0$

Example 1: Show that the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ are orthogonal.

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{bmatrix}, \text{ and } \mathbf{v}_3 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$$

Example 2: Are the following vectors, $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$ orthogonal?

Example 3: Given that a set of nonzero vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ are orthogonal, show that the set is linearly independent.

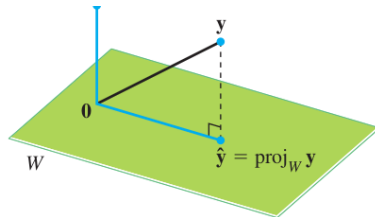
Orthogonal Projections

Given a nonzero vector $\mathbf{u} \in \mathbb{R}^n$, a different vector, $\mathbf{y} \in \mathbb{R}^n$ can always be decomposed into two components; one, a multiple of $\mathbf{u}$ (parallel to $\mathbf{u}$), and the orthogonal to $\mathbf{u}$.

The vector component of $\mathbf{y}$ that is a multiple of $\mathbf{u}$ or parallel to $\mathbf{u}$ is called the **Orthogonal Projection**: The projection of $\mathbf{y}$ onto $\mathbf{u}$ (if $\mathbf{u} \neq 0$) is:

$$\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}} \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

The vector, $\mathbf{y}$ can be written as a sum of $\mathbf{u}$ and $\text{proj}_{\mathbf{u}} \mathbf{y}$.



Alternatively if $W = \text{Span}(\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\})$ where the $\mathbf{u}_i$'s are orthogonal, we say the **orthogonal projection** of $\mathbf{y}$ onto the subspace spanned by W is

$$\hat{\mathbf{y}} = \text{proj}_{\mathbf{u}} \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \dots + \frac{\mathbf{y} \cdot \mathbf{u}_n}{\mathbf{u}_n \cdot \mathbf{u}_n} \mathbf{u}_n.$$

Examples

Example 1: Let $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$. Find the orthogonal projection of $\mathbf{u}_1$ onto $\mathbf{u}_2$. Then write $\mathbf{u}_1$ as the sum of two orthogonal vectors, one being $\text{proj}_{\mathbf{u}_2} \mathbf{u}_1$, and the other being the orthogonal complement.

Example 2: Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix} \right\}$, and W be the subspace spanned by $\mathcal{B}$. Let $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$.

Find the orthogonal projection of $\mathbf{v}$ onto the W .

Then write $\mathbf{v}$ as the sum of two orthogonal vectors, one being $\text{proj}_W \mathbf{v}$, and the other being the orthogonal complement.

Example 3: Let $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -2 \\ -5 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix} \right\}$, and W be the subspace spanned by $\mathcal{B}$.

Find a vector that is perpendicular to the the vectors in $\mathcal{B}$.

- i **Elementary Linear Algebra** by *David C. Lay*
- ii **Linear Algebra** by *Jim Hefferon*
- iii **Introduction to Linear Algebra** by *Gilbert Strang*
- iv **Linear Algebra with Applications** *W. Keith Nicholson*